

MS Algebra Recitation 5

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1 The Sylow Theorems

We'll state and prove the Sylow Theorems. We use the following lemma.

Lemma. *Let G be a finite group and $H \leq G$ a p -subgroup. Then $[G : H] \equiv [N_G(H) : H] \pmod{p}$.*

Proof. Act H on the set G/H of left cosets of H . The coset gH is a fixed point if and only if $hgH = gH$ for all $h \in H$, if and only if $g^{-1}hg \in H$ for all $h \in G$. Since H is finite, this happens if and only if $g^{-1}Hg = H$, if and only if $g \in N_G(H)$. Thus, the fixed points of the action are the cosets $N_G(H)/H$, and so there are $[N_G(H) : H]$ fixed points. Since H is a p -group, this means

$$[G : H] \equiv [N_G(H) : H] \pmod{p}.$$

■

This allows us to construct successively larger p -groups in finding Sylow p -subgroups:

Lemma. *Let G be a finite group and $H \leq G$ a p -subgroup with $p \mid [G : H]$. Then there is a $H \trianglelefteq K \leq G$ with $[K : H] = p$.*

Proof. We have that $0 \equiv [G : H] \equiv [N_G(H) : H] \pmod{p}$, so that p divides $[N_G(H) : H]$. By Cauchy's theorem, $N_G(H)/H$ has an element of order p , and thus a subgroup $1 \trianglelefteq \bar{K}$ of order p . This corresponds to a subgroup $H \trianglelefteq K \leq N_G(H) \leq G$ with $[K : H] = p$. ■

Finally, let's state and prove Sylow's Theorems.

Theorem (Sylow). *Let p be a prime and G be a finite group of order $p^r m$ with $p \nmid m$.*

- (a) G has a Sylow p -subgroup.
- (b) If $H \leq G$ is a p -subgroup and $P \leq G$ is a Sylow p -subgroup, then there is $g \in G$ such that $H \leq gHg^{-1}$. In particular, any two Sylow p -subgroups are conjugates.
- (c) $\# \text{Syl}_p(G) \mid m$ and $\# \text{Syl}_p(G) \equiv 1 \pmod{p}$.

Proof.

- (a) Let H be a p -subgroup of G of maximal order. If $|H| \neq p^r$, then $p \mid [G : H]$, so we can find $H \leq K \leq G$ with $[K : H] = p$. Then, $|K| = p|H|$ is a p -subgroup of G of strictly larger order, contradicting the maximality of H . Thus, $|H| = p^r$.
- (b) Let H act on the set G/P of left cosets of P . Since $|G/P| = [G : P] = m$ is not divisible by p , there must be at least one fixed point gP under the action of H . Now, $hgP = gP$ for all $h \in H$, so that $h \in gPg^{-1}$. Thus, $H \leq gPg^{-1}$.
- (c) Let G act on $\text{Syl}_p(G)$ by conjugation, and fix $P \in \text{Syl}_p(G)$. By (2) above, the orbit of P is $\text{Syl}_p(G)$. The stabilizer of P is $N_G(P)$, so $\# \text{Syl}_p(G) = [G : N_G(P)]$. Now,

$$m = [G : P] = [G : N_G(P)] \cdot [N_G(P) : P],$$

so $\# \text{Syl}_p(G) = [G : N_G(P)]$ divides m . Additionally, $m = [G : P] \equiv [N_G(P) : P] \pmod{p}$, so reducing the above equation modulo p and dividing by m gives $\# \text{Syl}_p(G) = [G : N_G(P)] \equiv 1 \pmod{p}$. ■

As an application of these theorems, we have the following:

Proposition (Frattini Argument). *Let $N \trianglelefteq G$ be finite and let P be a Sylow p -subgroup of N . Then $G = N_G(P)N$.*

Proof. Omitted. This ended up on the homework. ■

Note also that, if G has exactly one Sylow p -subgroup, then Sylow 2 implies that it is normal.

Example. Let's find how many Sylow 5-subgroups S_5 has. We know that this number n must be 1 (mod 5) and divide 24, so $n = 1$ or $n = 6$. We claim that $n = 1$.

It is a fact (to be proven later) that S_5 contains a subgroup A_5 of index 2, and that A_5 has no proper nontrivial normal subgroups. This means that the only normal subgroups of S_5 are $S_5, A_5, 1$. Indeed, if $N \trianglelefteq S_5$, then $N \cap A_5 \trianglelefteq A_5$. If $N \cap A_5 = A_5$ then $N = A_5$ or $N = S_5$ since $[S_5 : A_5] = 2$. If $N \cap A_5 = 1$, then $|NA_5| = |N| \cdot |A_5|$. Thus, $|N| = 1$ or $|N| = 2$. We know S_5 doesn't have any normal subgroups of order 2, so $N = 1$.

In particular, $n = 6$. So we have an action $S_5 \curvearrowright \text{Syl}_5(S_5)$ via conjugation, which gives us a map $S_5 \rightarrow S_6$. Since we know that S_5 moves each of its Sylow 5-subgroups to any other, this copy of S_5 inside S_6 is not one of the "obvious" copies $\{\sigma : \sigma(i) = i\}$. Note the map $S_5 \rightarrow S_6$ must have trivial kernel, because it doesn't have kernel A_5 or S_5 (its image has size ≥ 6).

2 Internal Direct Products

We'll study how to decompose groups in terms of their subgroups.

Lemma. *Let $H, N \trianglelefteq G$ with $H \cap N = 1$. Then elements of H commute with elements of N .*

Proof. Let $h \in H$ and $n \in N$. Then

$$[h, n] = h \underbrace{nh^{-1}n^{-1}}_{\in H} = \underbrace{hnh^{-1}}_{\in N} n^{-1}$$

lies in both H and N , and so must be equal to 1. That is, h and n commute. ■

So, we might hope that $HN \cong H \times N$. This is true.

Proposition. *Let G be a group with $H, N \trianglelefteq G$. Suppose that $G = HN$ and $H \cap N = 1$. Then $G \cong H \times N$.*

Proof. Our lemma shows that the map $H \times N \rightarrow G$ given by $(h, n) \mapsto hn$ is a homomorphism. It is clearly surjective. If $(h, n) \mapsto 1$, then $hn = 1$, so that $H \ni h = n^{-1} \in N$. Thus, we must have $h = n = 1$, and so this map is also injective, hence an isomorphism. ■

What if we choose to decompose with more groups? Unfortunately, pairwise intersections being trivial isn't enough. For instance, consider $V_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. The three groups $\langle(0, 1)\rangle$, $\langle(1, 0)\rangle$, and $\langle(1, 1)\rangle$ have pairwise trivial intersections, but clearly V_4 isn't the direct product of three copies of $\mathbb{Z}/2\mathbb{Z}$. We need a stronger condition (hopefully this reminds you of direct sums in linear algebra):

Proposition. *Let G a group, $H_1, \dots, H_k \trianglelefteq G$, and $G = H_1 \cdots H_k$. If $H_i \cap \prod_{j \neq i} H_j = 1$ for all $1 \leq i \leq k$, then $G \cong H_1 \times \cdots \times H_k$.*

Proof. We induct. $k = 1$ is trivial and $k = 2$ was done above. If $k \geq 3$, we have that $G = H_1(H_2 \cdots H_k)$ and $H_1 \cap H_2 \cdots H_k = 1$, so the $k = 2$ case shows $G \cong H_1 \times H_2 \cdots H_k$. Induction shows $H_2 \cdots H_k \cong H_2 \times \cdots \times H_k$. ■

As an example of how to use this, we have the following:

Proposition. *Let G be a finite group and suppose that, for each prime $p \mid |G|$, there is only one Sylow p -subgroup of G . Then G is isomorphic to the direct product of its Sylow p -subgroups.*

Proof. Write $|G| = p_1^{e_1} \cdots p_n^{e_n}$, and let P_i be the unique Sylow p_i -subgroup. Since we know all Sylow p_i -subgroups are conjugate, note that P_i is normal in G .

WLOG take $i = 1$ and consider $P_1 \cap \prod_{j \neq 1} P_j$. This is a subgroup of G , and so its order divides $|P_1| = p_1^{e_1}$ and $|\prod_{j \neq 1} P_j|$. The latter has order

$$\frac{|P_2|}{|P_2 \cap \prod_{j > 2} P_j|} \cdot \frac{|P_3|}{|P_3 \cap \prod_{j > 3} P_j|} \cdots \frac{|P_{n-1}|}{|P_{n-1} \cap P_n|} \cdot |P_n|$$

by iterating $|HK| = \frac{|H| \cdot |K|}{|H \cap K|}$. In particular, its order divides $|P_2| \cdot |P_3| \cdots |P_n| = p_2^{e_2} p_3^{e_3} \cdots p_n^{e_n}$. Since this is relatively prime to $p_1^{e_1}$, we see that $P_1 \cap \prod_{j \neq 1} P_j$ is trivial. This holds similarly for all i .

Finally, we need $G = P_1 \cdots P_n$. We'll look at the order of the latter:

$$\frac{|P_1|}{|P_1 \cap \prod_{j>1} P_j|} \cdot \frac{|P_2|}{|P_2 \cap \prod_{j>2} P_j|} \cdots \frac{|P_{n-1}|}{|P_{n-1} \cap P_n|} \cdot |P_n|$$

The same argument as above shows that each of the denominators must be 1 (since they each divide two relatively prime quantities), and so this has order $p_1^{e_1} \cdots p_n^{e_n} = |G|$. Thus, we have equality, and so our previous result gives $G \cong P_1 \times \cdots \times P_n$. ■